

Afirmación: Sea p primo, entonces

$\Phi_p(x) = x^{p-1} + x^{p-2} + \dots + x + 1$ es irreducible sobre $\mathbb{Q}$ (en $\mathbb{Q}[x]$)

Dem: Sea $x = t+1$.

$$f(t) = \Phi_p(x) = \Phi_p(t+1) = ?$$

$$\frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + x + 1, \text{ Así}$$

$$f(t) = \frac{(t+1)^p - 1}{t}$$

$$= \frac{t^p + \binom{p}{1}t^{p-1} + \dots + \binom{p}{p-1}t + 1 - 1}{t}$$

$$= t^{p-1} + \binom{p}{1}t^{p-2} + \dots + \binom{p}{p-2}t + p.$$

Eisenstein? Afirmación: $\forall 1 \leq k \leq p-1 \quad p \mid \binom{p}{k}$.

Si es verdadero, entonces $f(t)$ es irreducible y así $\Phi_p(x)$ es irreducible.

Prove que $p \mid \binom{p}{k}$ $1 \leq k \leq p-1$.

$$\binom{p}{k} = \frac{p!}{k! (p-k)!} \in \mathbb{Z}.$$

Si p es primo $(p, k!) = 1$, $(p, (p-k)!) = 1$.

así $(p, k! (p-k)!) = 1$.

Observe que $p! = \binom{p}{k} k! (p-k)!$

y $p \mid p!$ así $p \mid \binom{p}{k} k! (p-k)!$

luego necesariamente $p \mid \binom{p}{k}$

Teorema: Teorema Fundamental del Algebra

Sea $f(t) \in \mathbb{C}[t]$ no constante, entonces existe $\alpha \in \mathbb{C}$ tal que $f(\alpha) = 0$.

Definición: Multiplicidad

Sea $f(t) \in K[t]$, entonces $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{F}_p$.

- $\alpha \in K$ es raíz simple de $f(t)$ si $(t - \alpha) \mid f(t)$ y $(t - \alpha)^2 \nmid f(t)$.
- $\alpha \in K$ es raíz de multiplicidad m de $f(t)$ si $(t - \alpha)^m \mid f(t)$ y $(t - \alpha)^{m+1} \nmid f(t)$.

Teorema

El número de raíces de un polinomio sobre un subcuerpo de $\mathbb{C}$ (contando multiplicidades) es menor o igual a su grado.

Mark the following true or false. (Here 'polynomial' means 'polynomial over $\mathbb{C}$ '.)

- (a) Every polynomial of degree n has n distinct zeros. F
- (b) Every polynomial of degree n has at most n distinct zeros. $\checkmark$
- (c) Every polynomial of degree n has at least n distinct zeros. F
- (d) If f, g are non-zero polynomials and f divides g , then $\partial f < \partial g$. F
- (e) If f, g are non-zero polynomials and f divides g , then $\partial f \leq \partial g$. $\checkmark$
- (f) Every polynomial of degree 1 is irreducible. $\checkmark$
- (g) Every irreducible polynomial has prime degree. F $x^4 + 7$
- (h) If a polynomial f has integer coefficients and is irreducible over $\mathbb{Z}$, then it is irreducible over $\mathbb{Q}$. $\checkmark$
- (i) If a polynomial f has integer coefficients and is irreducible over $\mathbb{Z}$, then it is irreducible over $\mathbb{R}$. F
- (j) If a polynomial f has integer coefficients and is irreducible over $\mathbb{R}$, then it is irreducible over $\mathbb{Z}$. $\checkmark$

$$4t + 2 = 2(2t + 1) \text{ is reducible } / \mathbb{Z}.$$

Definición: Polinomios simétricos elementales

Los polinomios simétricos elementales en las n variables $x_1, x_2, \dots, x_n$ son:

$$\begin{aligned}e_1(x_1, x_2, \dots, x_n) &= \sum_i x_i \\e_2(x_1, x_2, \dots, x_n) &= \sum_{i < j} x_i x_j \\e_3(x_1, x_2, \dots, x_n) &= \sum_{i < j < k} x_i x_j x_k \\&\vdots \\e_n(x_1, x_2, \dots, x_n) &= x_1 x_2 \cdots x_n\end{aligned}$$

Teorema: (Fundamental de los polinomios simétricos)

Cualquier polinomio simétrico en n variables $x_1, x_2, \dots, x_n$ es representable de manera única como un polinomio en los polinomios simétricos elementales $e_1, e_2, \dots, e_n$.

Prueba (Gauss) (Idea ordenar los polinomios e "**bubble**" en su orden)
* orden Lexicográfico en $\mathbb{N}^n$

Casa y Carro.


Carro < Casa

$(1, 2, 3)$ vs $(1, 1, 5)$ vs $(1, 0, 3)$ vs $(2, 0, 0)$

$$(1, 0, 3) < (1, 1, 5) < (1, 2, 3) < (2, 0, 0).$$

multigrado. $m\partial(x_1^{i_1} \dots x_n^{i_n}) = (i_1, \dots, i_n)$.

$$m\partial(5y^3z^2) = (0, 3, 2)$$

$$m\partial(3x^2yz) = (2, 1, 1)$$

así diremos que $5y^3z^2 < 3x^2yz$.

multigrado de un polinomio en $x_1, x_2, \dots, x_n$

$$m\partial(f) = \max \{ m\partial(x_1^{i_1} \dots x_n^{i_n}) \mid \alpha x_1^{i_1} \dots x_n^{i_n} \text{ es un monomio de } f \}$$

Ej: $f(x, y, z) = 5y^3z^2 + 7xy^3z + 8x^2 + 3x^2yz$

$$m\partial(f) = ?$$

$$\{ (0, 3, 2), (1, 3, 1), (2, 0, 0), (2, 1, 1) \}$$

así $m\partial(f) = (2, 1, 1)$.

Ej: * Es un "orden total" en $\mathbb{R}[X_1, \dots, X_n]$

$$X^2 Y Z \text{ vs } 3X^2 Y Z.$$

$$* \text{m\partial}(f \cdot g) = \text{m\partial}(f) + \text{m\partial}(g).$$

* Toda cadena finita tiene m\u00edn.

Sea $f(X_1, \dots, X_n)$ sim\u00e9trico. y

$$\text{m\partial}(f) = (i_1, i_2, \dots, i_n) \Rightarrow i_1 \geq i_2 \geq \dots \geq i_n$$

Sim\u00e9trico significa $\forall \sigma \in S_n \quad f(X_1, \dots, X_n) = f(X_{\sigma(1)}, \dots, X_{\sigma(n)})$

$$\text{Ej: } f(x, y) = (x - y)^2 = x^2 - 2xy + y^2$$

$$\{(2, 0), (1, 1), (0, 2)\} \quad \text{m\partial}(f) = (2, 0)$$

Dem: Por Contradicci\u00f3n $\exists K;$

$$i_K < i_{K+1}$$

consider la permutación de intercambia

X_K y X_{K+1} . Así tenemos los monomios,

$$dX_1^{i_1} \dots X_K^{i_K} X_{K+1}^{i_{K+1}} \dots X_n^{i_n} \quad \text{y} \quad dX_1^{i_1} \dots X_K^{i_{K+1}} X_{K+1}^{i_K} \dots X_n^{i_n}$$

$\downarrow \qquad \qquad \qquad \downarrow$

$$(i_1, \dots, i_K, i_{K+1}, \dots, i_n) < (i_1, \dots, i_{K+1}, i_K, \dots, i_n)$$

Contradicción. Pues se sabe que el mayor es $(i_1, \dots, i_K, i_{K+1}, \dots, i_n)$.

Dem: (del Teo Fund de Polin. Simétrico) (Gauss)

Sea $f(x_1, \dots, x_n)$ simétrico y

$$m\partial(f) = (i_1, i_2, \dots, i_n)$$

entonces sea $g = \rho_1^{i_1-i_2} \rho_2^{i_2-i_3} \dots \rho_n^{i_n}$

así $m\partial(g) = m\partial(f)$.

obsérvese que

$$\begin{aligned} m\partial(\rho_1) &= (1, 0, \dots, 0) \\ m\partial(\rho_2) &= (1, 1, 0, \dots, 0) \\ m\partial(\rho_3) &= (1, 1, 1, \dots, 0) \\ &\vdots \\ m\partial(\rho_n) &= (1, 1, 1, \dots, 1) \end{aligned}$$

así $m\partial(\rho_1^{i_1-i_2} \rho_2^{i_2-i_3} \dots \rho_n^{i_n})$

$$= m\partial(\rho_1^{i_1-i_2}) + m\partial(\rho_2^{i_2-i_3}) + \dots + m\partial(\rho_n^{i_n})$$

$$= (i_1-i_2, 0, \dots, 0) + (i_2-i_3, i_2-i_3) + \dots + (i_n, \dots, i_n)$$

$$= (i_1, i_2, \dots, i_n)$$

$$\text{as } m\partial(f - dg) < m\partial(f).$$

$$\underline{f - h_1(\ell_1, \dots, \ell_n) - h_2(\ell_1, \dots, \ell_n) - \dots - h_k(\ell_1, \dots, \ell_n) = K.}$$

$$\varepsilon_f: f = x^4 + y^4 \quad \rho_1 = x+y \quad \rho_2 = xy$$

$$m\partial(f) = (4,0) \quad g = \rho_1^{4-0} \rho_2^0 = \rho_1^4$$

$$\begin{aligned} f_1 &= f - \rho_1^4 = x^4 + y^4 - (x+y)^4 \\ &= -4x^3y - 6x^2y^2 - 4xy^3 \end{aligned}$$

$$m\partial(f_1) = (3,1) \quad g_1 = \rho_1^{3-1} \rho_2^1 = \rho_1^2 \rho_2$$

$$\begin{aligned} f_2 &= f_1 - (-4\rho_1^2 \rho_2) = -4x^3y - 6x^2y^2 - 4xy^3 \\ &\quad + 4(x+y)^2xy \end{aligned}$$

$$f_2 = -\cancel{4x^3y} - 6x^2y^2 - \cancel{4xy^3} + \cancel{4x^3y} + 8x^2y^2 + \cancel{4xy^3}$$

$$f_2 = 2x^2y^2 \quad m\partial(f_2) = (2,2)$$

$$\text{Ans} \quad x^4 + y^4 = \rho_1^4 - 4\rho_1^2 \rho_2 + 2\rho_2^2$$

$$x^4 + y^4 + z^4.$$